

Design of a Sustainable Community-Based Item Exchange prototype

Ehda Altawfig

*Department of Computer Science
College of Science and Humanities
Imam Abdulrahman Bin Faisal
University*

Jubail, Saudi Arabia
2230003808@iau.edu.sa

Norah Mukhelib

*Department of Computer Science
College of Science and Humanities
Imam Abdulrahman Bin Faisal
University*

Jubail, Saudi Arabia
2230001721@iau.edu.sa

Rehab Alshehri

*Department of Computer Science
College of Science and Humanities
Imam Abdulrahman Bin Faisal
University*

Jubail, Saudi Arabia
2230005944@iau.edu.sa

Danah Aldbas

*Department of Computer Science
College of Science and Humanities
Imam Abdulrahman Bin Faisal
University*

Jubail, Saudi Arabia
2230006556@iau.edu.sa

Aysa Alrashed

*Department of Computer Science
College of Science and Humanities
Imam Abdulrahman Bin Faisal
University*

Jubail, Saudi Arabia
2230000218@iau.edu.sa

Safa Alsalem

*Department of Computer Science
College of Science and Humanities
Imam Abdulrahman Bin Faisal
University*

Jubail, Saudi Arabia
2230000274@iau.edu.sa

Layan Alharbi

*Department of Computer Science
College of Science and Humanities
Imam Abdulrahman Bin Faisal
University*

Jubail, Saudi Arabia
2230000924@iau.edu.sa

Abrar Alharbi

*Department of Computer Science
College of Science and Humanities
Imam Abdulrahman Bin Faisal
University*

Jubail, Saudi Arabia
2230003532@iau.edu.sa

D.Hanan k. Aljasim

*Department of Computer Science
College of Science and Humanities
Imam Abdulrahman Bin Faisal
University*

Jubail, Saudi Arabia
hkaljasim@iau.edu.sa

Abstract—SwapIt is a mobile application promoting sustainability through a non-commercial item exchange platform. Developed using Human-Computer Interaction (HCI) principles and a User-Centered Design (UCD) approach, the project addresses resource waste by facilitating community-based reuse. The design process involved task analysis, prototyping, and rigorous usability evaluation, including heuristic evaluation and cognitive walkthroughs. While initial results confirmed a simple and navigable interface, testing identified gaps in system feedback and help visibility. These issues were resolved by integrating error messages, confirmation screens, and a notification center. Ultimately, SwapIt demonstrates a usable, trust-based solution for sustainable consumption and community engagement.

Keywords—Human-Computer Interaction, Sustainability, User-Centered Design, Item Exchange, Sharing Economy, Usability Testing.

I. INTRODUCTION

In recent years, sustainability and responsible consumption have become major concerns due to the increasing amount of waste and overconsumption of resources. According to the World Bank, global waste generation is expected to increase by 70% by 2050 unless urgent action is taken [1]. Many usable items are discarded although other people may still benefit from them. This problem creates unnecessary environmental impact and reduces opportunities for reuse. Technology can help address this issue by supporting community-based sharing and

encouraging users to exchange items instead of purchasing new ones.

SwapIt is a mobile design created to promote sustainability by enabling users to exchange unused items within a community-based platform. Unlike traditional marketplace designs that mainly focus on buying and selling, SwapIt focuses on item exchange, reuse, and social responsibility. The design allows users to list items they no longer need, browse items offered by others, send exchange requests, communicate within the system, and rate users after completing exchanges.

From a Human-Computer Interaction perspective, SwapIt emphasizes usability, accessibility, and simplicity. The interface is designed to reduce cognitive load, provide clear navigation, and support users with different technical backgrounds. The main functionalities of the system include user registration, profile creation, item listing, categorized browsing, keyword-based searching, exchange requests, in-app messaging, notifications, rating and feedback, and a user-friendly interface.

The project follows a User-Centered Design approach. This means that the design decisions were based on user needs, task requirements, usability principles, and evaluation results. During the project, the team developed paper prototypes, refined the interface, created a high-fidelity prototype using Figma, and evaluated the design through different usability testing methods. These stages helped identify usability issues and improve the final design.

The objective of this paper is to present SwapIt as a sustainable community-based item exchange design developed using Human-Computer Interaction principles. The paper discusses the background of the project, related literature, design direction, usability findings, and final project outcomes. By combining sustainability goals with user-centered design, SwapIt aims to provide a practical platform that encourages responsible consumption, community collaboration, and trust-based item exchange.

II. LITERATURE REVIEW

In recent years, sustainability and responsible consumption have become important global concerns due to the increasing amount of waste and overconsumption of resources. Many usable items are discarded even though they can still benefit others, which negatively affects both the environment and society. This behavior increases environmental pollution, wastes valuable resources, and encourages unnecessary consumerism. As communities continue to rely heavily on fast consumption and constant purchasing, the need for sustainable digital solutions has become more important than ever.

Technology and digital platforms now play a significant role in supporting sustainable behavior by encouraging reuse, reducing waste, and strengthening community collaboration. As a result, researchers have explored how sharing economy systems and community-based digital platforms can create positive environmental and social impact [2]. These studies became the foundation for the development of the SwapIt prototype design, which aims to provide a simple and sustainable way for users to exchange unused items within their communities.

III. Sustainability and Responsible Consumption

Sustainability is the core concept behind the SwapIt prototype design. Previous studies explained that sharing economy platforms can support sustainable development by improving resource efficiency and extending the lifecycle of products [3]. Instead of throwing away unused items, users can redistribute them to others who may still benefit from them. This idea directly supports the purpose of SwapIt, which focuses on reducing waste through item reuse and community exchange.

Research also showed that digital systems can encourage sustainable behavior through social interaction, feedback, and user motivation [4]. This influenced the design direction of SwapIt by encouraging features such as notifications, ratings, and future reward systems that motivate users to continue exchanging items responsibly.

In addition, studies highlighted the importance of localized exchange systems that support reuse within communities [5]. This concept is reflected in SwapIt because the prototype design focuses on helping users exchange items within nearby communities in a simple and convenient way. At the same time, some studies warned that sharing systems may unintentionally encourage unnecessary accumulation if users collect items without real need [6]. Because of this, SwapIt focuses on meaningful exchange through organized categories, clear item descriptions, and transparent communication between users.

IV. Sharing Economy and User Participation

Several existing systems helped shape the direction of SwapIt by showing both the strengths and limitations of current sharing platforms. Applications such as OLX [7], Facebook Marketplace [8], and Freecycle [9] provide users with features such as browsing, messaging, and item exchange. However, most of these systems mainly focus on commercial buying and selling rather than sustainability and responsible consumption.

Research related to collaborative consumption explained that users are motivated not only by environmental awareness, but also by practical benefits such as saving money and accessing useful items [10]. Other studies also identified social interaction, technological convenience, and economic benefit as major reasons behind user participation in sharing platforms [11], [12].

These findings influenced the SwapIt prototype design by emphasizing simplicity, accessibility, and community interaction. Unlike commercial systems, SwapIt focuses specifically on non-monetary exchange and sustainable community participation. The goal is not only to help users exchange items, but also to encourage a more responsible and collaborative lifestyle.

V. Trust, Safety, and Usability

Trust is one of the most important factors in any platform where users interact with unfamiliar individuals. Studies showed that reputation systems, ratings, and feedback mechanisms improve accountability and increase user confidence in digital platforms [13], [14]. Without trust, users may hesitate to exchange items or communicate with others online.

Research also emphasized that users feel safer when platforms provide transparent communication, clear profiles, and understandable interaction processes [15]. For this reason, the SwapIt prototype design includes ratings, reviews, notifications, and feedback systems that help users feel more comfortable during the exchange process.

Another important issue discussed in the literature is usability and user experience. Studies explained that clear feedback, notifications, and confirmation messages reduce confusion and improve interaction quality [16]. This became highly important during the usability evaluation of SwapIt because early testing revealed that users felt uncertain during some actions due to missing feedback and unclear navigation. As a result, the design was improved by adding confirmation screens, notifications, clearer error messages, and more visible help features.

Privacy and security were also identified as important concerns in digital sharing systems [17]. Therefore, the SwapIt prototype design focuses on secure communication and controlled interaction to reduce risk and improve user trust.

VI. Community Engagement and Long-Term Participation

The success of community-based platforms depends on continuous participation and user engagement. Studies

showed that digital systems become more successful when users actively communicate, share information, and participate in community activities [18], [19], [20]. Features such as messaging, reviews, notifications, and feedback systems increase long-term interaction and strengthen relationships between users [21], [22].

Research also highlighted the importance of organized digital structures that make browsing and item management easier for users [23], [24]. This influenced the categorized browsing and organized item listing system in SwapIt, making the prototype design easier to navigate and more user-friendly.

In addition, some studies discussed how gamification features such as reward points, badges, and achievement systems can motivate users and make sustainable actions feel more rewarding [25]. This idea supports future improvements planned for SwapIt, where sustainability-related rewards and engagement systems can encourage users to remain active within the platform.

Overall, the reviewed literature demonstrated that successful sharing platforms must combine sustainability, usability, trust, security, and community engagement. Based on these findings, the importance of the SwapIt prototype design became clearer because the project provides not only a technical solution, but also a socially and environmentally meaningful contribution. These studies helped show the value of the project and strengthened the idea of creating a sustainable and community-centered platform that encourages responsible consumption and item reuse.

VII. METHODOLOGY & MATERIALS

4.1 Research Design Approach

This study adopts a User-Centered Design (UCD) methodology to guide the development and evaluation of the proposed system, SwapIt. UCD was selected because it ensures that the system is designed based on real user needs, behaviors, and usability requirements rather than assumptions.

The methodology follows an iterative design cycle consisting of stages such as Understanding the context of use, Specifying user requirements, Designing and prototyping and Evaluating and refining the system

This iterative process allowed continuous improvement of the system based on user feedback and usability principles.

4.2 Data Collection Methods

4.2.1 Survey

A structured online survey was distributed through a combination of university networks and social media platforms in order to reach a diverse range of potential users aged 18 and above. The survey was designed to capture attitudes and behaviors relevant to the design of the proposed system, covering five main areas: general interest in sustainability and item exchange, willingness to use swapping platforms, perceived practicality of swapping as an alternative to purchasing, comfort level when exchanging items with other people, and concerns related to trust and safety. Additional questions explored whether participants felt that swapping influenced their broader behavior toward reusing items, and whether they would consider continuing to use swapping as part of their everyday lifestyle.

A total of 92 responses were collected. All participants confirmed that they were 18 years of age or older and provided informed consent before completing the survey.

The responses indicated strong overall support for the concept of item swapping. 89.1% of respondents agreed that swapping contributes to sustainability and waste reduction, and 79.3% considered it a practical alternative to purchasing new items, suggesting that both environmental motivation and everyday usefulness play an important role for this user group. Looking further ahead, 71.7% indicated that they would consider continuing to use item swapping as part of their lifestyle, which points to potential for sustained engagement rather than one-time use.

Despite these positive attitudes, the results also revealed a clear concern around trust: 57.6% of respondents reported feeling hesitant or unsure about exchanging items with strangers. This suggests that while users embrace the environmental and practical value of swapping, interpersonal trust remains a meaningful barrier to adoption.

These findings directly influenced the inclusion of features such as rating systems, verified profiles, and secure in-app communication in the system design, which were prioritized in order to address the trust-related concerns identified through the survey.

4.2.2 Interviews

Semi-structured interviews were conducted to gain deeper qualitative insights into user behavior and expectations. The interviews were designed as a comparative evaluation between the existing application “Badalha” and the proposed system “SwapIt”.

A total of 18 participants from the general public took part, with ages ranging from 19 to 39 years and an equal split of 9 male and 9 female respondents. About 44% had previously used Badalha, while the rest evaluated it for the first time. Interviews were conducted in a mixed setting, with some sessions held face-to-face and others online.

The interview guide was organized into three parts: questions about participants’ experience with Badalha (first impressions, ease of use, navigation, search, uploading, and safety); a comparison with SwapIt after introducing the concept; and a 1-to-5 rating of both applications across:

- Ease of use
- Security
- Motivation
- Clarity

The findings showed that Badalha was perceived as simple and easy to register on, but weak in filtering, item descriptions, and trust due to the absence of a verification system. Most participants preferred the proposed system because it appeared easier to use, better organized, and more engaging, and they rated it higher than Badalha across all four criteria. The reward and badge system was seen as a strong motivator, while ratings and reviews were considered important for building trust. The most frequently suggested improvements were verified profiles, better filtering, and integrated in-app communication.

These qualitative findings directly informed the inclusion of user verification, ratings and reviews, in-app

communication, and gamification elements in the final design of SwapIt.

4.3 Design Process

After completing the user and task analyses, the design was developed in stages, with each version reviewed and improved before moving on to the next. The process followed five main steps:

1. Initial paper prototypes: Low-fidelity sketches were drawn for the main screens of the application — Home, Login, Create Account, Product, Profile, Chat, and Rating. At this stage, the focus was on the overall layout and the flow between screens, not on visual styling.
2. Evaluation against design principles: The initial prototypes were reviewed against established interface design principles, which highlighted issues in consistency, feedback, navigation, and cognitive load.
3. Design refinement: The issues identified in the review were then addressed through a series of improvements: a consistent bottom navigation bar, unified button styles and a single-color theme, simpler and less cluttered layouts, clearer labels and feedback after user actions, and a more organized arrangement of content across screens.
4. Final paper prototypes: The refined screens were brought together into a final set of paper prototypes, which captured the agreed layout and interactions.
5. High-fidelity prototype: The final paper prototypes were then built out as a high-fidelity prototype in Figma, with basic interactions added to simulate the real user flow.

4.4 Evaluation Methods

To evaluate the usability and effectiveness of the proposed system, multiple evaluation techniques were conducted

4.4.1 Heuristic Evaluation

The system was evaluated by 8 designers using Nielsen's usability heuristics, including:

- Visibility of system status
- Consistency and standards
- Error prevention
- User control and freedom
- Recognition rather than recall

The evaluation identified several usability issues, such as unclear feedback messages, inconsistent elements, and limited error prevention features.

4.4.2 Usability Testing

Several usability testing methods were conducted throughout the project to evaluate the effectiveness, usability, and interaction quality of the proposed system. The testing process focused on identifying usability issues, understanding user behavior, and improving the overall user experience.

Discount Usability Testing was conducted to evaluate basic user tasks such as login, registration, browsing products, and adding items for exchange. Participants were asked to complete these tasks while the team observed interaction

difficulties, validation problems, and feedback issues. This testing helped identify unclear buttons, missing confirmation messages, and navigation problems that affected task completion.

Cognitive Walkthrough testing was conducted to evaluate whether users could understand and complete tasks independently without external guidance. The testing focused on task flow clarity, interface understanding, navigation simplicity, and interaction predictability. Participants were asked to perform tasks step-by-step while evaluators analyzed whether users could easily determine the next action required by the system.

Remote Usability Testing was also conducted using participants' personal devices in online settings. Participants tested the complete application flow, including browsing items, searching, chatting, requesting swaps, completing exchanges, and submitting ratings and reviews. This testing evaluated the overall user experience, interaction quality, communication efficiency, ease of navigation, and user satisfaction in realistic usage environments.

The usability testing results revealed several important usability issues related to feedback visibility, navigation clarity, trust mechanisms, and interaction flow. Based on these findings, multiple improvements were implemented throughout the system. These improvements included adding confirmation and error messages, improving navigation flow, enhancing notification systems, improving trust and verification features, organizing categories and favorites more clearly, and improving visual clarity and interaction feedback.

These refinements significantly improved the usability, consistency, and overall user satisfaction of the proposed system.

4.5 Tools and Technologies

Several tools and technologies were used throughout the project development process. Figma was used to design the high-fidelity interactive prototype and create the user interface screens. Google Forms was used to distribute the survey and collect quantitative data from participants. WhatsApp and face-to-face meetings were used to conduct interviews and communicate with participants during the data collection process. Zoom and online communication platforms were also used to conduct remote usability testing sessions and team collaboration. Microsoft Word was used for documentation and report writing throughout the project.

4.6 Ethical Considerations

Participants were informed about the purpose of the study before participating in surveys and interviews. Consent was obtained from all participants, and all collected information was treated confidentially. No sensitive personal information was collected or shared during the research process.

VIII. RESULTS

Paper prototypes of the "SwapIt" application were made during the initial design phase, with an emphasis on the basic system structure and user interactions. After testing these prototypes, important areas for improvement were identified, such as improving the navigation flow, guaranteeing visual

consistency, and adding notifications for user actions. Significant changes were made, including the implementation of a bottom navigation bar, unified color schemes and button styles, clear error messages, and the addition of a notifications screen to improve transparency. By improving control and clarity in interactions, these modifications produced a final design that was far more user-friendly.

Following these improvements, the final high-fidelity prototype was developed using Figma. The prototype represents the complete and refined version of the SwapIt application, incorporating all usability enhancements identified during the evaluation phase. The figures below showcase illustrative examples of the application's key screens and interactions, each carefully refined based on the usability findings to ensure a more intuitive, consistent, and user-friendly interaction flow.

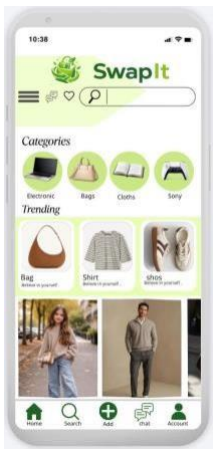


Fig. 1. SwapIt Home Screen

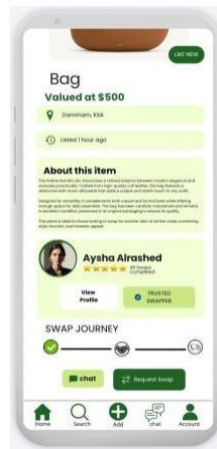


Fig. 2. Item Details and Swap

Fig. 1. This figure illustrates the primary interface for the discovery phase, featuring a categorized navigation system and a trending section that simplifies the user's search for potential swap items.

Fig. 2. This screen details the core task of item evaluation, where the user can view the item's estimated value and location, integrated with the "Swap Journey" feature to track the progress of the exchange.

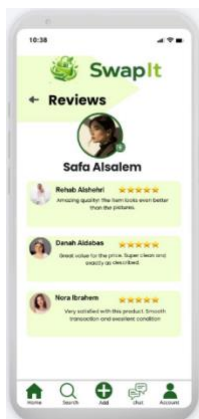


Fig. 3. User Reviews Interface Screen

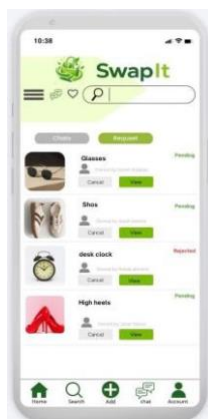


Fig. 4. Request Management Screen

Fig. 3. This interface represents the social proof mechanism within the application, displaying comprehensive user reviews and ratings to establish transparency and build trust between swappers in the community.

Fig. 4. This figure demonstrates the request management system, enabling users to efficiently organize their interactions by monitoring the real-time status of pending, accepted, or rejected swap offers.



Fig. 5. My Swaps Screen

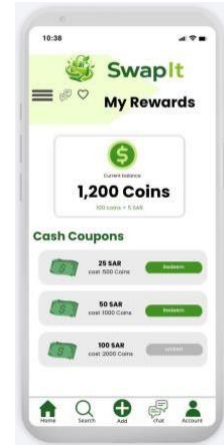


Fig. 6. My Rewards Screen

Fig. 5. This screen provides a structured overview of the user's personal activity, categorizing previous transactions into active or completed swaps to facilitate historical record-keeping.

Fig. 6. This figure showcases the gamification and incentive model of the application, displaying the rewards and coins earned through sustainable swapping behavior to encourage long-term user retention.

IX. DISCUSSION

The refinement of the SwapIt final prototype addresses the critical gap between environmental willingness and user hesitation. Although 79.3% of survey participants recognized item swapping as a practical alternative to purchasing, 57.6% expressed deep concerns regarding transacting with unfamiliar peers. Our analysis revealed that this hesitation stems from the "information vacuum" and operational ambiguity found in current peer-to-peer designs. During initial testing, users experienced noticeable cognitive friction due to the lack of immediate feedback on their actions. Consequently, our design strategy transitioned to an actively communicative system that replaces uncertainty with direct, real time status updates, ensuring high predictability and user confidence.

This user feedback directly shaped the architectural layout of the Request Management interface Fig. 4. To resolve violations of Nielsen's "Visibility of System Status" heuristic, the interface was re-engineered to deliver definitive, color-coded state indicators specifically Pending and Rejected. Additionally, displaying clear ownership accountability, such as indicating items "Owned by Danah Aldabas" or "Owned by Sarah Jenkins", directly satisfies the user's need for transaction validation.

Providing immediate tactical actions through the integrated "Cancel" and "View" buttons for specific items (like Glasses, Shoes, desk clock, and High heels) grants users enhanced autonomy, fulfilling the HCI mandate for user control and freedom.

Finally, comparing SwapIt to the existing local application Badalha [26] highlights its significant functional improvements. Comparative interviews indicated that users perceived Badalha as weak due to its lack of organized filtering and verification. SwapIt counteracts these limitations by introducing verified profiles, peer-reviewed ratings Fig.3 and a gamified framework. Testing of the "Reward and Badge System" Fig. 6 confirmed that earning 1,200 Coins and redeeming Cash Coupons act as critical behavioral motivators rather than mere aesthetics. Ultimately, combining real-time system feedback with robust trust mechanisms successfully eliminates psychological barriers, allowing localized sustainable platforms to compete with commercial markets.

IX. FUTURE WORK

In future development, SwapIt will be implemented as a complete application, and the possible errors that may occur during real use will be explored and improved. The application will be tested to identify issues related to user registration, item listing, swap requests, communication between users, and system performance. This will help improve the reliability, usability, and overall quality of the application.

Artificial Intelligence can be added in the future to improve SwapIt by increasing security, supporting users through an AI chat assistant, improving search results, and providing personalized item recommendations based on user activity and preferences.

X. CONCLUSION

SwapIt is a sustainable community-based item exchange application designed to encourage reuse, reduce waste, and support responsible consumption. The project applied Human-Computer Interaction principles to create a simple, clear, and user-centered mobile interface. The application focuses on important user needs such as easy item listing, browsing and searching, exchange requests, in-app messaging, notifications, ratings, and feedback.

Throughout the project, the design was developed and improved through several stages, including requirement analysis, comparison with similar systems, user analysis, task analysis, paper prototyping, high-fidelity prototyping, heuristic evaluation, cognitive walkthroughs, and usability testing. These stages helped identify both strengths and usability issues in the application.

The evaluation results showed that SwapIt has several strengths, including a simple interface, clear labels, smooth navigation, logical task flow, and visually engaging product displays. At the same time, some issues were identified, such as unclear feedback, missing confirmation messages, inactive buttons, and limited visibility of help features. These issues

were addressed through improvements such as adding error messages, confirmation screens, a notifications page, clearer help text, and improved navigation.

In conclusion, SwapIt demonstrates how HCI principles can be applied to design a socially responsible application. By focusing on usability, trust, sustainability, and community engagement, the application can help users exchange items more safely and efficiently while contributing to waste reduction and responsible consumption.

REFERENCES

- [1] World Bank, "What a Waste 2.0: A Global Snapshot of Solid Waste Management to 2050," World Bank Group, Washington, DC, 2018.
- [2] M. J. Cohen, "The future of consumer society: Prospects for sustainability in the new economy," Oxford University Press, 2020.
- [3] A. Boar, "A systematic literature review. Relations between sharing economy, sustainability and sustainable development goals," in XVIII Congreso Internacional de Investigadores en Economía Social y Cooperativa, Barcelona, Spain, 2020.
- [4] O. Mosca, A. Manunza, S. Manca, G. Vivanet, and F. Fornara, "Digital technologies for behavioral change in sustainability domains: a systematic mapping review," *Frontiers in Psychology*, vol. 14, Art. no. 1234349, Jan. 2024. doi: 10.3389/fpsyg.2023.1234349.
- [5] M. Björklund, S. Gustafsson, and K. Skill, "Sustainability potentials of digitally based platforms for the circularity of household items," *Cleaner Logistics and Supply Chain*, vol. 10, Art. no. 100133, 2024. doi: 10.1016/j.clscn.2023.100133.
- [6] B. Parguel, R. Lunardo, and F. Benoit-Moreau, "Sustainability of the Sharing Economy in Question: When Second-Hand Peer-to-Peer Platforms Stimulate Indulgent Consumption," working paper, 2016.
- [7] OLX Group, "OLX Online Classifieds Platform," [Online]. Available: <https://www.olx.com>. [Accessed: 14-May-2026].
- [8] Meta Platforms, Inc., "Facebook Marketplace: Buy and Sell Locally," [Online]. Available: <https://www.facebook.com/marketplace>. [Accessed: 14-May-2026].
- [9] The Freecycle Network, "Changing the world one gift at a time," [Online]. Available: <https://www.freecycle.org>. [Accessed: 14-May-2026].
- [10] T. Luo, S. S. Kanhere, J. Huang, S. K. Das, and F. Wu, "Sustainable Incentives for Mobile Crowdsensing: Trust and Reputation Systems," *IEEE*, 2017.
- [11] P. Rojanakit, S. de Oliveira, and K. J. McCarthy, "A critical review and research agenda for the sharing economy," *Journal of Business Research*, vol. 139, pp. 1164-1185, 2022.
- [12] J. Hamari, M. Sjöklint, and A. Ukkonen, "The sharing economy: Why people participate in collaborative consumption," *Journal of the Association for Information Science and Technology*, vol. 67, no. 9, pp. 2047-2059, 2016.
- [13] N. A. Alwi, S. S. S. Mahmood, and N. M. Nawi, "Drivers and motivation factors of the sharing economy: A systematic literature review," *Cogent Business & Management*, vol. 8, no. 1, 1997245, 2021.
- [14] C. Gullberg and N. Gustafsson, "Accountability through (Inter)Action? A Framework of Affordances for Understanding Civil Society Accountability on Social Media Platforms," *Social Media + Society*, vol. 11, no. 2, pp. 1-14, 2025.
- [15] A. Al-Adwan, A. S. Al-Adwan, and J. Berger, "Sharing economy: A systematic review of definitions, drivers, applications, industry status, and business models," *ResearchGate*, 2022.
- [16] Y. Zhang and X. Chen, "Impact of the sharing economy on sustainable development: A systematic literature review," *Sustainability*, vol. 16, no. 4, 1702, 2024.
- [17] A. Moilanen, "The Role of Trust in the Sharing Economy: What factors are influencing consumer trust in peer-to-peer platforms," Master's thesis, Tampere University of Applied Sciences, Nov. 2023.
- [18] R. H. Pereira, M. J. A. Gonçalves, and M. A. G. Coelho, "Reputation Systems: A framework for attacks and frauds classification," *J. Inf. Syst. Eng. Manag.*, vol. 8, no. 1, 19218, 2023.

- [19] N. Johnson, M. L. Druckenmiller, F. Danielsen, and P. L. Pulsifer, "The use of digital platforms for community-based monitoring," *BioScience*, vol. 71, no. 5, pp. 452–466, May 2021.
- [20] N. M. V, M. S. A, and N. P. Selvaraj, "Next-Step: A mobile-based platform for community-driven issue reporting and resolution," *SRM Institute of Science and Technology*, India, 2024.
- [21] B. Bayraktar, K. Ragupathi, and K. A. Troyer, "Building Trust Through Feedback: A Conceptual Framework," *Teaching & Learning Inquiry*, vol. 13, pp. 1–19, 2025.
- [22] J. Yao, Y. Zhang, and R. Wang, "Beyond expectation and confirmation: Unraveling the drivers of continued engagement with Chinese short video apps," *Telematics and Informatics Reports*, vol. 18, 100209, 2025.
- [23] T.-D. T. Pham, F.-Y. Lo, and K.-H. Huarng, "Unmasking privacy apprehension: A bibliometric review of mobile sharing economy applications," *J. Innov. Knowl.*, vol. 10, no. 1, 100656, 2025.
- [24] M. N. B. Razali and W. H. Wan Ishak, "A User-centered Digital Repository: Designing and Evaluating HobbyHaven for Collectors," *Journal of Digital System Development (JDSD)*, vol.3, no. 1, pp. 68–76, April 2025.
- [25] S. Marth, B. Hartl, and E. Penz, "Sharing on platforms: Reducing perceived risk for peer-to-peer platform consumers through trust-building and regulation," *J. Consum. Behav.*, vol. 21, no. 6, pp. 1255–1267, 2022.
- [26] Baddlha. .Android (Google Play):
[\[https://play.google.com/store/apps/details?id=com.sbtechnology.baddlha\]](https://play.google.com/store/apps/details?id=com.sbtechnology.baddlha)(<https://play.google.com/store/apps/details?id=com.sbtechnology.baddlha>)